

SHIMOFUSA AB, Japan

WMO: 477270

Lat: 35.799N

Lon: 140.012E

Elev: 108

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.4	30.2	8.6	8.6	39.7	10.8	9.6	40.6	21.2	46.3	18.5	45.3	3.4	310	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	92.9	79.0	89.9	77.7	87.9	77.3	80.1	88.2	79.2	86.4	78.5	85.2	8.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	148.0	85.1	77.2	141.9	83.7	76.7	139.5	83.2	43.9	89.0	42.9	86.8	42.0	85.5	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	17.5	15.2	24.4	96.9	2.2	2.8	22.9	98.9	21.6	100.5	20.3	102.0	18.7	104.1
			21.5	82.4	1.8	1.3	20.3	83.3	19.2	84.0	18.2	84.7	17.0	85.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	40.6	42.2	47.9	57.4	65.7	71.4	78.7	81.0	74.3	64.3	53.8	44.6
	DBStd	14.73	3.85	4.64	5.52	5.95	4.91	4.63	5.26	4.63	5.80	5.15	5.41	4.82
	HDD50	836	292	225	109	10	0	0	0	0	0	0	22	178
	HDD65	3262	756	640	529	236	50	6	0	0	4	74	336	631
	CDD50	4581	1	5	45	231	486	642	889	961	728	444	137	12
	CDD65	1533	0	0	0	7	72	198	424	496	282	53	1	0
	CDH74	12175	0	0	0	26	316	972	3761	4957	1948	194	1	0
	CDH80	4095	0	0	0	1	44	195	1370	1919	542	24	0	0
	WSAvg	6.6	5.6	6.4	7.2	8.0	7.8	7.0	7.6	7.5	6.8	5.8	4.9	5.0
	PrecAvg	57.2	2.6	2.6	4.2	4.5	5.1	5.9	5.2	4.8	7.9	8.0	4.1	2.4
(15) Precipitation	PrecMax	77.2	5.8	5.0	7.6	8.1	8.4	11.1	11.0	11.3	15.7	23.6	9.8	7.6
	PrecMin	41.0	0.0	0.4	1.7	1.7	2.5	1.7	0.6	0.6	1.4	1.0	0.3	0.1
	PrecStd	8.2	1.7	1.3	1.5	1.8	1.6	2.0	3.1	3.1	3.9	5.3	2.2	1.5
	PrecStd	8.2	1.7	1.3	1.5	1.8	1.6	2.0	3.1	3.1	3.9	5.3	2.2	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.3	64.5	69.8	77.2	84.5	89.2	95.3	96.7	91.5	82.8	71.9	64.5
		MCWB	50.8	54.1	56.0	61.4	65.7	74.0	79.0	79.9	76.7	70.1	61.5	55.7
	2%	DB	55.4	58.8	64.8	73.5	80.4	84.5	91.8	93.3	88.1	78.8	68.4	59.4
		MCWB	45.6	49.7	53.6	60.2	65.4	72.2	78.5	79.5	76.6	68.7	59.8	50.2
	5%	DB	52.0	55.1	62.3	70.2	77.2	82.2	89.7	91.2	85.9	75.5	66.2	57.0
		MCWB	41.7	45.4	53.3	59.4	64.8	70.9	77.6	78.6	76.1	66.7	57.5	48.1
	10%	DB	49.8	51.8	59.1	68.0	75.0	79.2	87.6	89.2	82.7	73.2	63.0	53.8
		MCWB	40.2	42.5	50.5	58.1	64.1	69.9	77.3	78.2	74.4	65.2	55.8	45.7
	0.4%	WB	53.3	57.9	60.0	68.2	72.5	77.4	81.4	82.3	79.5	74.4	65.5	59.0
		MCDB	57.8	62.4	64.1	71.0	76.7	85.0	91.2	92.5	86.4	78.6	69.3	62.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.4	51.1	57.5	64.3	69.3	75.5	79.8	80.7	78.5	71.4	62.3	52.9
		MCDB	52.3	56.6	62.9	70.3	74.8	80.8	88.3	89.0	84.4	76.7	66.1	57.8
	5%	WB	43.7	46.9	54.9	61.9	67.6	74.0	78.9	79.7	77.4	69.0	60.0	49.7
		MCDB	50.0	51.9	60.5	67.5	73.6	78.9	86.5	87.2	83.0	73.7	64.1	54.3
	10%	WB	41.3	43.7	51.3	59.8	66.0	72.2	78.0	79.0	76.3	66.9	57.8	46.9
		MCDB	47.1	49.6	57.4	65.6	72.4	77.3	85.0	85.8	81.6	71.6	61.7	51.9
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	16.6	16.9	16.8	15.2	12.8	12.7	12.9	12.4	13.3	15.2	16.4
		MCDBR	20.8	21.8	20.9	19.9	19.1	17.0	16.5	15.9	14.6	16.4	18.3	19.7
		MCWBR	13.8	15.0	14.0	11.0	8.6	7.3	6.3	5.7	6.3	9.2	11.9	13.4
	5% WB	MCDBR	18.6	19.3	18.4	16.1	15.3	13.2	14.7	14.1	12.9	13.6	15.8	16.7
		MCWBR	14.1	15.3	15.0	11.0	9.1	7.2	6.2	6.0	6.4	9.2	12.2	13.4
(40) Clear-Sky Solar Irradiance	taub		0.379	0.428	0.498	0.548	0.569	0.603	0.595	0.581	0.513	0.482	0.426	0.382
	taud		2.238	2.092	1.929	1.852	1.858	1.841	1.908	1.932	2.083	2.097	2.168	2.249
	Ebn at Noon		252	251	244	239	236	227	228	228	237	231	232	242
	Edh at Noon		34	44	56	64	65	66	61	59	48	43	36	32
(44) All-Sky Solar Radiation	RadAvg		865	1025	1286	1502	1617	1440	1556	1564	1213	954	787	755
	RadStd		58	101	80	145	167	129	259	151	105	96	72	57

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page